

Open System-One Decision Models: Frozen Diffusion-LM Backbones and Specialist Decision Heads for Calibrated Scientific Decisions

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Abstract

Structured decisions—selecting among options, verifying whether evidence supports a claim, and grading response quality on ordinal scales—are the atomic operations of automated scientific pipelines. Proprietary “System-One” models deliver such decisions as typed API calls without generation, but are closed. We present an open, trainable counterpart: a frozen masked-diffusion language model backbone (LLaDA-8B) with small specialist decision heads producing probabilities for three decision types (choice, noul, score). We contribute three findings. (1) A deterministic canonical ordering of options yields exact permutation invariance (option-order flip rate $0.77 \rightarrow 0.00$), where train-time logit averaging does not. (2) Multi-layer attention pooling over option spans outperforms last-layer mean pooling for choice (+7 points), while an ordinal auxiliary loss improves graded scoring (+9 points). (3) Scaling training data $30\times$ on an out-of-domain corpus improves in-distribution accuracy to 0.96 but does not transfer (−13 points on the target domain), showing that coverage of the target reasoning distribution—not volume—is the binding constraint. On public benchmarks, OPEN-JEV reaches 0.853 on SciFact claim verification and transfers zero-shot to general classification (0.930 SST-2, 0.854 AG News), while parametric knowledge tasks remain bounded by the backbone ceiling (0.315 GPQA vs. backbone ≈ 0.33).

1 Introduction

[TODO: intro: typed decisions vs generation; routing to System Two; contributions list]

2 Setup

Decision types. We consider three typed decisions over a state (context) and a question: **choice** selects one of K answer options; **noul** decides whether evidence supports a claim ($K=2$); **score** assigns an ordinal grade $0 < 1 < 2$ (unrelated / related-but-wrong / correct).

Model. Each example is encoded as a single masked sequence: context, question, and options separated by marker tokens. A frozen masked-diffusion backbone (LLaDA-8B-Instruct) produces token representations; a specialist head per decision type maps pooled option-span features to class logits. Two head architectures are compared: a mean-pooled MLP and *AttnPool*, a learned query attending over each option span on concatenated hidden states from layers $\{8, 16, 24, 32\}$. Only the heads are trained ($\sim 2\text{M}$ parameters); the backbone remains frozen.

Canonical ordering. Options are sorted by token-id content before encoding, and gold indices remapped. The presentation order is thus a pure function of the option *set*: any input permutation collapses to the same sequence and permutation invariance holds exactly, at training, calibration, and evaluation.

Ordinal auxiliary loss. For score we add a CORAL-style term $\sum_{j=1}^{K-1} \text{BCE}(P(y \geq j), \mathbf{1}[g \geq j])$ on cumulative softmax mass, respecting the grade order without changing the K -way decode.

Metrics. Accuracy, expected calibration error (ECE), option-order flip rate (accuracy change under input permutation), and *automation@5%err*: the fraction of decisions that can be auto-accepted while keeping realized error $\leq 5\%$ —the operating metric for a decision layer.

3 Experiments

3.1 Scientific decision battery

[TODO: describe elite battery + v2 corpus (qa_raw, 25K pids, 3 corpora)]

3.2 Public benchmarks

GPQA main (448 graduate-level questions) and diamond subset (198) as choice; SciFact dev (340 claim–abstract pairs) as noul and 3-way score; plus general classification (SST-2, AG News, Enron spam, Banking77) to probe out-of-domain transfer. Benchmarks are eval-only; temperature is fitted on the battery dev split (no benchmark leakage).

4 Results

4.1 Ablations on the scientific battery

	choice	noul	score	flip (choice)
Baseline (mixed MLP)	0.775	0.707	0.522	0.76
+ orders= 4	0.721	0.782	0.586	–
+ AttnPool	0.847	0.636	0.588	0.75
+ both	0.749	0.780	0.576	0.70
Per-type heads	0.876	0.710	0.574	0.77
Per-type + canon. + ord.	0.870	0.783	0.608	0.00

Table 1: Ablation on the scientific battery (frozen LLaDA-8B). Canonical ordering eliminates order sensitivity exactly; per-type specialist heads plus AttnPool give the best choice accuracy and automation.

Key findings: (1) train-time logit averaging over permutations does *not* fix order sensitivity and costs choice accuracy; (2) canonical ordering achieves flip rate exactly 0 at no accuracy cost; (3) AttnPool on intermediate layers helps choice (+7) but hurts noul, motivating per-type heads; (4) the ordinal auxiliary loss lifts score $0.522 \rightarrow 0.608$.

	in-distribution (v2)	elite (target)
choice	0.965	0.738
noul	0.942	0.782
score	–	0.616

Table 2: Heads retrained on $30\times$ more data from a different QA distribution (qa_raw corpora) reach near-ceiling in-distribution accuracy but do not transfer to the elite battery, whose questions stress causal/multihop/negation reasoning absent from the source corpus.

4.2 Data scaling does not transfer across domains

4.3 Public benchmarks

benchmark	type	OPEN-JEV	reference
GPQA main	choice	0.315	GPT-4 ~ 0.39 ; expert ~ 0.65 ; backbone ceiling ~ 0.33
GPQA diamond	choice	0.328	
SciFact	noul	0.853	
SciFact	score	0.624	
Enron spam	choice	0.752	published System-One: 0.987
SST-2	choice	0.930	
AG News	choice	0.854	
Banking77 ($K=77$)	choice	0.229	

Table 3: Zero-shot transfer to public benchmarks (eval-only; temperature fitted on the battery dev split, no benchmark data seen). Published System-One numbers are third-party measurements (Kumar, 2026) on identical public datasets. GPQA accuracy sits at the reported backbone ceiling (LLaDA-8B ≈ 0.33), indicating the readout already extracts the backbone’s accessible parametric knowledge. Transfer to $K \leq 4$ classification is strong (SST-2, AG News); $K=77$ degrades because training used $K \leq 4$.

5 Discussion

[TODO: what a System-One layer buys: determinism, calibration, automation budget; routing uncertain cases (auto@5%); the domain-coverage lesson; relation to published System-One results (third-party reports only).]

6 Limitations

Backbone pretraining may include benchmark-adjacent text (standard public benchmark caveat). Our model is trained on one scientific QA distribution; cross-domain transfer is limited by construction. Score remains the weakest decision type for all systems studied, including the proprietary reference.

References

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